EXERCISE 04

Flavio Massimiliano Ricci
ricci.2083804@studenti.uniroma1.it

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1 Introduction

Solution of Exercise04.

2 Methodology

2.1 Parametric and iterative solutions

The parametric curve is $(\alpha(\epsilon), \epsilon)$, with $\alpha(\epsilon) = (1-\epsilon)\pi\cot(\pi\epsilon)$ for $0 < \epsilon \leq 1/2$. For $\alpha > 0$, rearranging gives

$$f_1(\epsilon, \alpha) = 1 - \frac{\alpha}{\pi}\tan(\pi\epsilon), \qquad f_2(\epsilon, \alpha) = \frac{1}{\pi}\arctan\left(\frac{\pi(1-\epsilon)}{\alpha}\right). \tag{1}$$

Starting at ϵ_0, we update $\epsilon_{i+1} = \epsilon_i + a\Delta_i$, where $\Delta_i = f_j(\epsilon_i, \alpha) - \epsilon_i$. A run succeeds if a valid iterate reaches $|\Delta_i| < 10^{-10}$ within 10^3 updates. The solver rejects non-finite iterates, values outside $[0, 1/2]$, and the singular endpoint $1/2$ for f_1.

2.2 Convergence

By the mean value theorem, $\Delta_{i+1} = [1-a+af_j'(\xi_i, \alpha)]\Delta_i$, where ξ_i lies between successive iterates and the prime denotes the ϵ derivative.
The condition $|1 - a + af_j'(\epsilon^*, \alpha)| < 1$ guarantees local convergence to the fixed point ϵ^*, but not global convergence.
By the Banach fixed-point theorem, convergence from any initial value in a closed interval I is guaranteed if the iterative map sends I into itself and is a contraction throughout I.
For $f_1' = -\alpha\sec^2(\pi\epsilon)$ this gives $a < 2/[1 + \alpha\sec^2(\pi\epsilon^*)]$, therefore not all a values are possible.
Moreover, when ϵ approaches $1/2$ the tangent can also send iterates outside the physical interval (to $-\infty$), so the convergence interval has an upper bound smaller than $1/2$.

For $0 < a \leq 1$, the f_2 map preserves the interval $[0, 1/2]$, because $0 < f_2 < 1/2$ and the map is therefore a convex combination. On this interval,

$$|f_2'| = \frac{\alpha}{\alpha^2 + \pi^2(1-\epsilon)^2} \leq \frac{1}{2\pi(1-\epsilon)} \leq \frac{1}{\pi}. \tag{2}$$

Thus $|1 - a + af_2'| \leq 1 - a + a/\pi < 1$ uniformly.
By the mean value theorem, for any $x, y \in I = [0, 1/2]$,

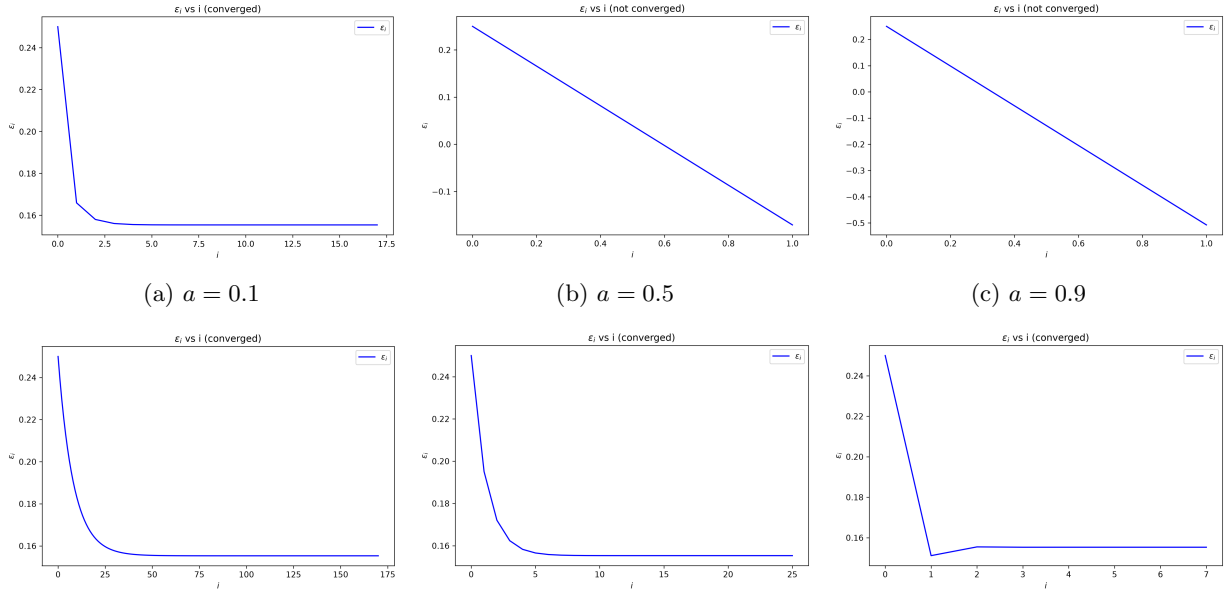
$$|T(x) - T(y)| \leq \left(1 - a + \frac{a}{\pi}\right) |x - y|,$$

where $T(\epsilon) = (1-a)\epsilon + af_2(\epsilon, \alpha)$.
Therefore, the hypotheses of the Banach fixed-point theorem are satisfied, ensuring the existence and convergence to the unique fixed point from any $\epsilon_0 \in [0, 1/2]$.

3 Results

3.1 Iteration traces

At $\alpha = 5$ and $\epsilon_0 = 0.25$, f_1 converges for $a = 0.1$. For $a = 0.5$ and 0.9, the fixed point is repelling and the first iterate is already negative. All three f_2 runs converge in 170, 25 and 7 updates, respectively, as a increases.



(a) $a = 0.1$ (b) $a = 0.5$ (c) $a = 0.9$

(d) $a = 0.1$ (e) $a = 0.5$ (f) $a = 0.9$

Figure 1: Iterates for f_1 (top) and f_2 (bottom) at $\alpha = 5$ and $\epsilon_0 = 0.25$.

3.2 Convergence heatmaps

For each map we test a uniform grid of 100 values of $\alpha \in [0.1, 10]$, 10 values of $\epsilon_0 \in [0.01, 0.5]$ and 10 values of $a \in [0.1, 0.9]$. Each cell at (ϵ_0, a) shows the number of successful runs out of 100, divided by 100. Over the whole grid, f_2 succeeds in every run and f_1 in 15.4%.

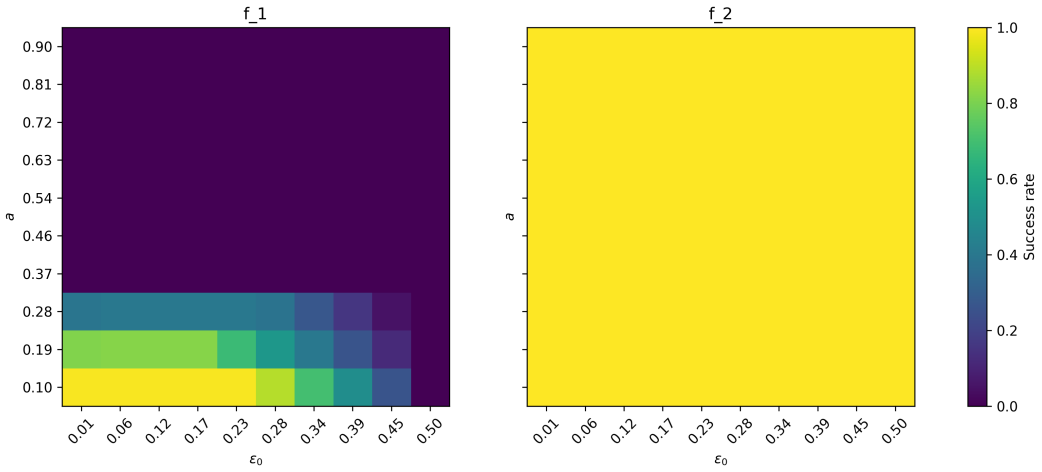
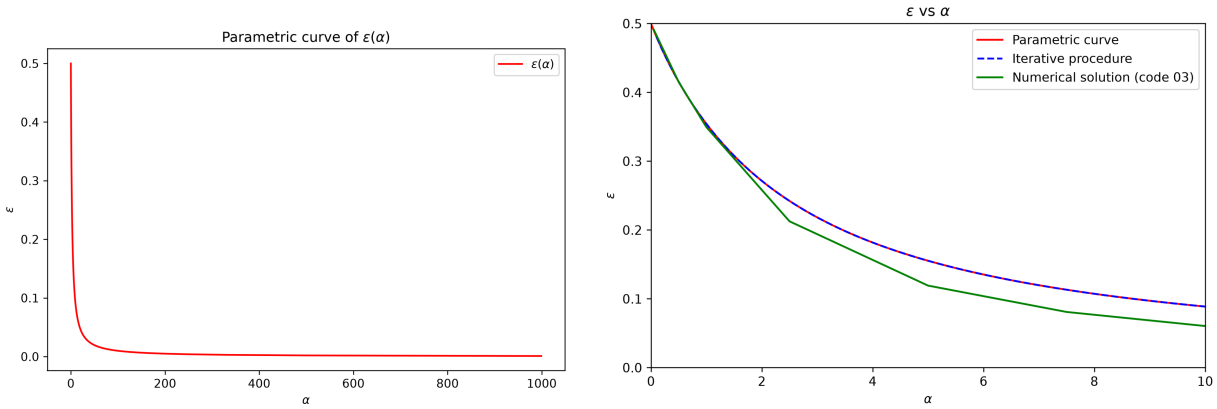


Figure 2: Success fraction over α for each (ϵ_0, a).

3.3 Automatic procedure and parametric curve

We rank the tested configurations by decreasing success rate, then by increasing mean residual and mean iteration count. The latter uses successful runs. This selects f_2 with $(\epsilon_0, a) = (0.5, 0.9)$. The solver restarts from $\epsilon_0 = 0.5$ at each of the 100 positive α values. Every run reaches the 10^{-10} residual tolerance.



(a) Parametric curve.

(b) Comparison over $0 \leq \alpha \leq 10$.

Figure 3: Annealed learning curve and comparison with the Exercise 03 simulation at $N = 20$.

4 Conclusions

The damped f_2 map converges across the entire grid. Its learning curve overlaps the parametric curve, while the Exercise 03 estimate lies below both at larger α.